

Gyeongsu (Bob) Cho

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Profile

I am an integrated M.S.-Ph.D. student in Artificial Intelligence at UNIST, working on 3D/4D vision and generative modeling under the supervision of Prof. Kyungdon Joo. My research centers on animatable animal and human reconstruction from monocular images and videos, with recent work including DogRecon (IJCV 2025), AniGauss (CVPR 2026 Computer Vision for Animals Workshop), and WildAni4D (CVPR 2026 Findings). I am interested in building practical and scalable 3D/4D content generation systems through synthetic data, geometry-aware learning, and generative models.

Education

Ulsan National Institute of Science and Technology (UNIST) Integrated M.S.–Ph.D., Artificial Intelligence Research focus: 3D/4D vision, generative models, 3D content creation	2022 – Present
Chung-Ang University (CAU) Bachelor of Engineering, Mechanical Engineering (Early Graduation)	2018 – 2021

Research Interests

3D/4D reconstruction of animals and humans, generative models, Gaussian splatting, and video understanding.

Publications

Journal Articles

[J1]. **DogRecon: Canine Prior-Guided Animatable 3D Gaussian Dog Reconstruction From a Single Image**
Gyeongsu Cho, C. Kang, D. Soon, K. Joo. *International Journal of Computer Vision (IJCV)*, 2025.

Conference Papers

- [C1]. **Multi Dance Project**
Submitted to *ECCV*; currently under review.
- [C2]. **Reference-Guided Video Anomaly Detection for Display Inspection**
Submitted to *IROS*; currently under review.
- [C3]. **AniGauss: Toward Animatable Animal Reconstruction from Single In-the-Wild Images via Topology-Aware Gaussians**
Gyeongsu Cho, Changwoo Kang, Donghyeon Soon, Hezhen Hu, Kyungdon Joo. *CVPR 2026, Computer Vision for Animals Workshop*.
- [C4]. **WildAni4D: Towards 4D Animal Mesh Reconstruction**
Gyeongsu Cho, H. Hu, D. Soon, C. Kang, K. Joo. *CVPR 2026 Findings*.
- [C5]. **Pose-Diverse Multi-View Virtual Try-on from a Single Frontal Image**
S. Han*, M. Chung*, Gyeongsu Cho, K. Joo, T. Kim. *WACV*, 2026.
- [C6]. **Avatar++: Fast and Pose-Controllable 3D Human Avatar Generation from a Single Image**
S. Han*, M. Chung*, Gyeongsu Cho, K. Joo, T. Kim. *ICCV 2025, Wild3D Workshop*.
- [C7]. **Canine Prior-Guided Animatable 3D Gaussian Dog Reconstruction From a Single Image**
Gyeongsu Cho, C. Kang, D. Soon, K. Joo. *CVPR 2024, Computer Vision for Animals Workshop (Oral)*.
- [C8]. **SlaBins: Fisheye Depth Estimation using Slanted Bins on Road Environments**
J. Lee, Gyeongsu Cho*, J. Park*, K. Kim*, S. Lee*, J. H. Kim, S.-G. Jeong, K. Joo. *ICCV*, 2023.

Research & Industry Experience

Graduate Researcher, UNIST – 3D/4D Vision & Graphics	2022 – Present
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- Lead author of **DogRecon (IJCV 2025)**, **AniGauss (CVPR 2026 Computer Vision for Animals Workshop)**, and **WildAni4D (CVPR 2026 Findings)** on single-image and 4D animatable animal reconstruction.

- Develop synthetic video pipelines, SMAL-based animal models, and PyTorch3D-based renderers for large-scale training and evaluation.
- Explore generative pipelines that lift text/video outputs into animatable 3D/4D representations.

Research Collaboration with Samsung Electronics – Video Anomaly Detection 2024 – 2025

- Co-develop a video anomaly detection framework for industrial display inspection, using real manufacturing data.
- Contributed to dataset design, model architecture, and experiments for an **IROS** submission on reference-guided video anomaly detection.

Research Collaboration with UT Austin – 4D Animal and Human Reconstruction 2025 – Present

- Ongoing collaboration with Dr. Hezhen Hu (Postdoctoral Fellow, UT Austin) on 4D animal and human reconstruction research.
- Co-develop research projects including **WildAni4D**, **AniGauss**, and an **ECCV**-submitted multi-dance project.
- Work on model design and research development for animatable 4D animal and human motion and mesh reconstruction from monocular videos.

Research Collaboration with 42dot – Autonomous Driving Research (Hyundai Motor Group) 2022 – 2023

- Collaborated on autonomous driving research with 42dot, the autonomous driving division of Hyundai Motor Group.
- Contributed to a project accepted to **ICCV 2023**.

Startup Project – Real-Time 3D Human Pose Estimation (KIC & GWU Program) 2024

- Built a **3-camera real-time 3D human pose estimation system** with smoothing algorithms for stable motion analysis in gyms.
- Designed the system as an AI-based personal training assistant to support.
- Selected for a Korean government startup support program and a 3-week entrepreneurship program at **George Washington University**.
- Conducted 50+ interviews with fitness and healthcare professionals across major U.S. cities and refined a US-market-focused business model; ranked **1st in track**.

Teaching & Mentoring

- Teaching Assistant: Linear Algebra (CAU), Computer Graphics (UNIST), Introduction to AI Programming 2 (UNIST).
- Tutor for a Hyundai Motor Company internal data/AI program (Python, data analysis).

Grants & Awards

AI Tech Open Workshop – Creative Project UNIST (Startup Track, 1st Place) 2025

- Project: *Lightweight camera system and inference for 3D animal model reconstruction*.
- Awarded **30,000,000 KRW (20,000 USD)**; funding allocated as research budget for continued 3D animal reconstruction work.

Patents

- **Method and apparatus for reconstructing a 3D dog Gaussian model from a single image**
Korean Patent Application 10-2025-0107831 (2025-08-05); PCT Application PCT/KR2025/019381.
- **Method and apparatus for generating a depth map of an image**
Korean Patent Application 10-2023-0125078, with 42dot.

Reference

Prof. Kyungdon Joo, Professor, UNIST

Relationship: M.S & Ph.D. advisor

E-mail: kyungdon@unist.ac.kr

Dr. Hezhen Hu, Postdoctoral fellow, UT Austin

Relationship: Collaborator

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